ADITYA UNIVERSITY

Freshman Engineering Department

Department of Mathematics

Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-III: Partial Differential Equations.

Solve the following Equations (From 1 to 16):

1. $p\tan x + q\tan y = \tan z.$
2. $\frac{y^2z}{x}p + xzq = y^2.$
3. $y^2p - xyq = x(z - 2y).$
4. $x^2(y - z)p + y^2(z - x)q = z^2(x - y).$
5. $px^2 - qy^2 = z(x - y).$
6. $(z^2 - 2yz - y^2)p + (xy + zx)q = (xy - zx).$
7. $(mz - ny)p + (nx - lz)q = ly - mx.$
8. $(x^2 - y^2 - z^2)p + 2xyq = 2xz.$
9. $4r + 12s + 9t = 0.$
10. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}.$
11. $(D^2 - DD')z = \sin x \cos 2y.$
12. $(D^3 + D^2D' - DD'^2 - D'^3)z = 3\sin(x + y).$
13. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x + y.$
14. $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2y.$
15. $(D^2 - DD' - 2D'^2)z = (y - 1)e^x.$
16. $(D^2 + 2DD' - 8D'^2)z = \sqrt{2x + 3y}.$

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